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Experiment 5: M2M Communications (under IoT Scenario) in AVR Processors

1 AIM

This experiment deals with the M2M communications (under IoT Scenario) in AVR processors.

1. Demonstrate M2M communication of two AVR-IoT boards through MQTT protocol using Mosquitto broker (server)
2. AVR-IoT board A publishes and AVR-IoT board B subscribes.
3. Board B connected to a LCD module and displays (a) message sent by Board A and (b) light sensor data published (sent) by Board A.

2 Equipments, Hardware Required

To perform this experiment, the following components are required.

1. AVR-IoT WG boards - 2

2. All neccessary cables (USB, power cables etc) and LCDs to be connected with the subscriber board B for displaying received data

3. Laptop or desktop for configuring it as a mosquitto broker

4. Download free softwares: MQTT libraries, mosquito software and MPLAB IDE platform.

3 Main Concept

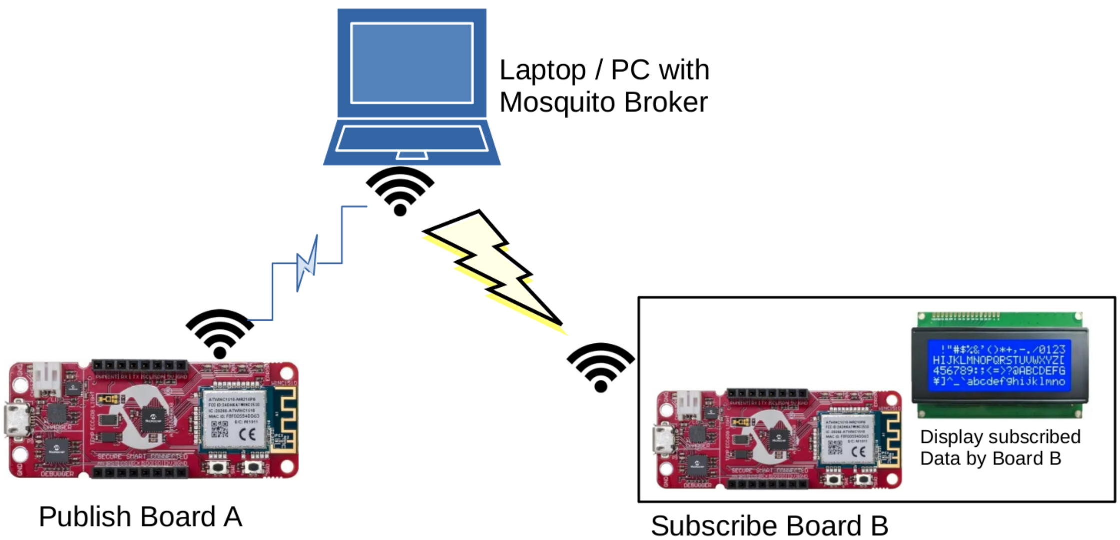
M2M communication stands for machine-to-machine communications. It is to be noted that a machine (IoT here) when certain parameters exceed a preset threshold (at random time instant), the machine (IoT) itself without manual intervention, initiates the conversation with another machine (IoT, server, actuator etc) and the communication ensues. This communication among machines without the manual intervention is called M2M communications.

[Alternatively, a machine (IoT) can continuously transmit a physical / engineering quantity to another machine which could process it and takes a decision].

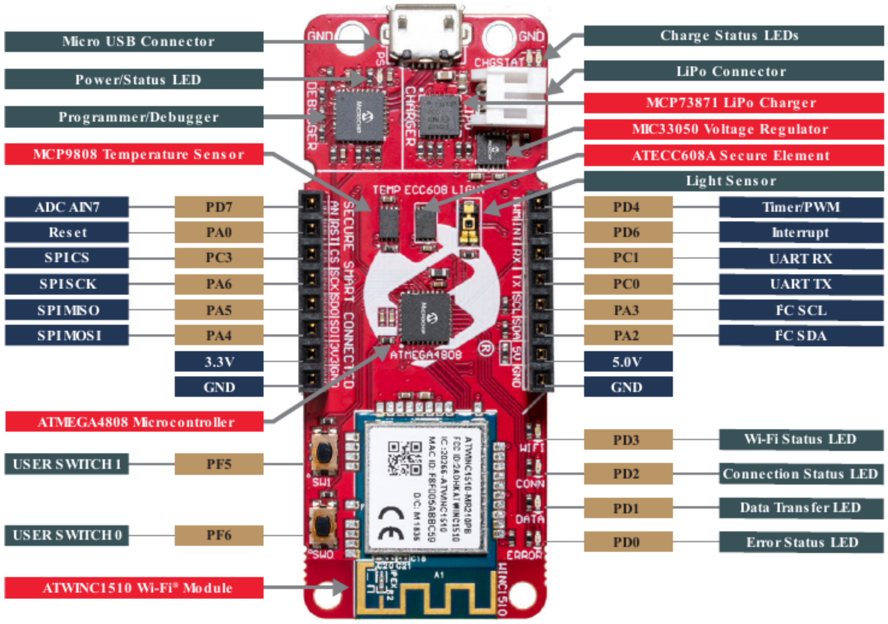
4 Methodology

4.1 Hardware Connections for M2M Communications

Powering the AVR-IoT boards can be done by connecting through USB cable. Board A (Publish) is connected to laptop (which is configured as 'Mosquitto' broker), wirelessly (through WiFi). (See subsection , on how to connect through WiFi). Similarly, Board B (Subscribe) is connected to the Mosquitto broker through WiFi. Since, the received data at Board B (Subscribe) is to be displayed in an LCD, LCD is connected to Board B (Subscribe) as given below. [Note: Board A is NOT connected to Board B DIRECTLY].



The only hardware connection intricacies are at the subscribing Board B. The correct pins of the board B has to be connected to the correct LCD pins.



Make the connections according to the connection diagram of the LCD with AVR-IoT board, as below. <Include, here a picture of connection diagrams for AVR-IoT board to LCD terminals?>

AVR Microcontroller Pin	**LCD Screen Pin**	**Connection**	
PA0 (RS)	RS	Control Pin (Register Select)	
Ground	RW	Control Pin (Read/Write, connect to Ground)	
PC3 (E)	E	Control Pin (Enable)	
PA4 (D4)	D4	Data Pin (4-bit mode)	
PA5 (D5)	D5	Data Pin (4-bit mode)	
PA6 (D6)	D6	Data Pin (4-bit mode)	
PD7 (D7)	D7	Data Pin (4-bit mode)	
+5V	VDD	Power Supply (5V)	
Ground	VSS	Ground	
Potentiometer Wiper	V0/VEE	Contrast Adjustment	

4.2 M2M Communications: MQTT Implementation along with Mosquitto Server (Broker)

The problem in this experiment is to implement M2M communications between two IoTs, through an intermediate broker. One board 'publish'es (Board A) on a default topic through the mosquito server (a laptop), while the other '*subscribes*' (Board B) to the above default topic (after connecting to the above mosquito broker through WiFi) and receives the message. In our experiment, the 'publish'ed message is displayed on the LCD connected to Board B. We also intend to display the light sensor data 'publish'ed (or transmitted) by Board A, in the LCD, connected to Board B. The Board B (Subscribe) independently publishes on EE2016IITM and is not monitored.

Following is the procedure for connecting two AVR-IoT boards using MQTT with a Mosquitto server to realize above M2M communication [McroChip1].

4.2.1 Outline of Procedure:

Each topic mentioned below is elaborated in later sections.

Mosquitto server: Install Mosquitto server on your laptop . Instructions for various operating systems can be found on the Mosquitto website (https://mosquitto.org/). We recommend that your laptop be installed the mosquitto broker (Connect https://mosquitto.org/ through iitmwifi, download and install mosquitto).

AVR-IoT boards: Ensure your boards have Wi-Fi connectivity and are programmed with an MQTT library compatible with the AVR architecture, with Board A burnt with publish code (mainPublish.c supportPublish.c codes uploaded in moodle) and Board B with subscribe code (mainSubscribe.c supportSubscribe.c uploaded in moodle). The above publish & subscribe functions are tweaked according to our requirement for our experiment. The whole process of configuring the AVR-IoT boards as publisher / subscriber is simplified, if you use Microchip's MPLAB IPE. (Hence, download and install Microchip's MPLAB IPE in your labtop). In the next subsection, we give step-by-step procedure for configuring the AVR-IoT boards as publisher / subscriber. The main step is that in MPLAB IPE, our edited code (supportPublish.c & supportSubscribe.c uploaded in moodle) is copied into the MPLAB window and the code is generated which is burnt in the respective AVR-IoT boards and run it simultaneously with the mosquitto broker ON. The supportPublish.c & supportSubscribe.c are edited, commented versions of /Source Files/MCC_Generated Files/examples/mqtt_example.c These edited ones incorporate the requirements of the experiment (i.e., to display the received message or received light sensor data in LCD display).

[Alternatively, you can also download, the publish independently tweak mqtt_example.c and use them to solve the problem posed. But, it is time consuming and strenuous exercise. We instead reccommend supportPublish.c to copy in place of /Source Files/MCC_Generated Files/examples/mqtt_example.c file in the same directory in MPLAB environment [MPLAB1] and ditto for Board B (Subscribe). Dump the program into respective boards and run it simultaneously with the mosquitto broker ON].

Demonstration (Tasks): (1) Every 'publish' cycle (default, once every 10 sec), the LED should blink, repeate the experiment by changing the publish rate and LED blink rate would increase, (2) receive the published message (say, "Hello World"), through subscription and display it in LCD and (3) publish, receive and display in LCD the light sensor data, sensed at the publish node. EDIT

MPLAB IPE: For all of the above, download and install Microchip's MPLAB IPE in your laptop (ubuntu versions are also available). It does lot things, which makes things easier for you. Create a new project on IoT (details given below), download and include MQTT codes corresponding to the publish and subscribe functionalities of the IoT nodes, edit the mqtt_example.c in the MPLAB IPE (edited versions are supportPublish.c). The original mqtt_example.c is replaced with its edited version and this would be used to burn codes into the respective AVR-IOT WG boards (but after this, boards are disconnected and independently run, ideally as they are battery operated, emulating the real-world IoT scenario).

4.2.2 Steps:

Step 1: Set up Mosquitto Server (on your own laptop or desktop in IE lab): Here we describe, how to set up the AVR® IoT WG development board as a Message Queuing Telemetry

Transport (MQTT) client [McroChip1]. For this, we can use more applications, like MyMQTT (android app), MQTT Explorer (desktop application) and Wireshark (a free and open-source packet analyzer) which are described in next subsection.

Since, the M2M communication we intend to implement is brokered (as opposed to peer-to-peer communication), we adopt mosquitto broker. In what follows, we give the procedure for installing the mosquitto broker [Msqtto].

1. Go to the Mosquitto web site.

2. In the Windows section, download Mosquitto (usually 64-bit version) and execute the EXE file to install it. (Alternatively, for ubuntu go to Linux distributions section).

3. Mosquitto Config (See [McroChip1])

   (a) Go to the file where you installed it and right-click on mosquitto.config then select Open with > Notepad++. That is the file we will execute. All lines are commented with "#" for now, so it is empty. Go to the end of the file, write, and save the following code [MPLAB1] :

   listener 9001

   protocol websockets

   listener 1883

   allow_anonymous true

   password_file C:\Program Files\mosquitto\pwfile.example

   - 1st line opens port 9001 as listener.
   - The 2nd line establishes port 9001 will use WebSocket protocol.
   - The 3rd line opens port 1883 as listener.
   - The 4th line allows connection to clients without a registered username (i.e., it's not in #pwfile.example).
   - The 5th line sets pwfile.example as the file that contains what usernames are registered, and what are the passwords for each of them.

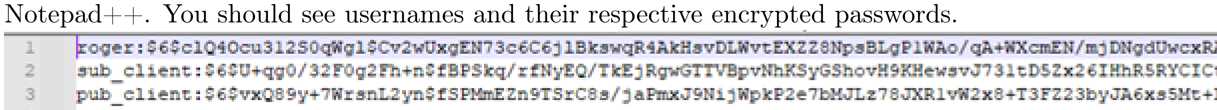
     A common mistake in this code is to add comments or spaces in the code. In this program language, every character is considered. Also, be careful not to add any spaces in the code. If you make a comment, it will need to be the whole line. Do not make a comment after a coding line since the program will not understand it and it will fail.

   The broker will open 9001 for WebSockets, and port 1883, where no username or password is needed. You can write "allow_anonymous false" instead of "allow_anonymous true", so only users with identified usernames and passwords will be allowed. We will see the utility of pwfile.example in the next section.

   An example (pwfile.example) is given in [McroChip1].

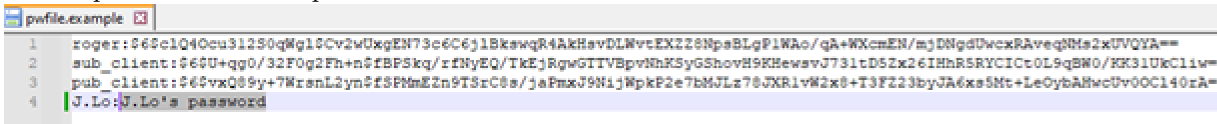
   (b) pwfile.example

      i. Go to Mosquitto's location. Right-click on pwfile.example, then select Open with > Notepad++. You should see usernames and their respective encrypted passwords.



```
roger:$6$clQ4Ocu312S0qWgl$Cv2wUxgEN73c6C6jlBkswqR4AkHsvDLWvtEXZZ8NpsBLgPlWAo/qA+WXcmEN/mjDNgdUwcxR
sub_client:$6$U+qg0/32F0g2Fh+n$fBPSkq/rfNyEQ/TkEjRgwGTTVBpvNhKSyGShovH9KHewsvJ73ltD5Zx26IHhR5RYCIC
pub_client:$6$vxQ89y+7WrsnL2yn$fSPMmEZn9TSrC8s/jaPmxJ9NijWpkP2e7bMJLz78JXRlvW2x8+T3FZ23byJA6xs5Mt+
```

ii. By default, "roger", "sub_client" and "pub_client" are the allowed usernames. The password for each of them is "password", but it is encrypted. Delete those lines or add a new username/password. For example, here the added username is "J.Lo", and its password is J.Lo's password.



iii. Save the file, close it and your Mosquitto broker should be able to connect using other usernames/passwords.

4. Execute Mosquitto as a Broker

   Initialize Mosquitto:

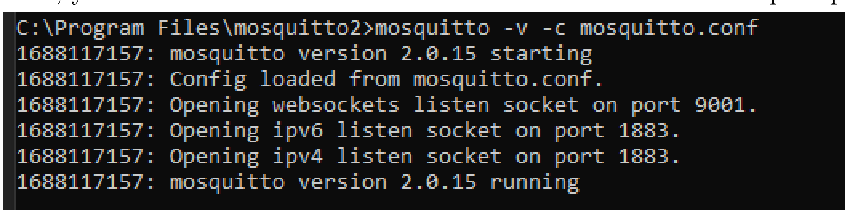
   (a) Open Command Prompt on your computer. Usually, this can be done by clicking on the Windows key and typing "Command Prompt".

   (b) Open the directory where mosquitto has been installed. Usually, that can be done with the text line: cd C:\Program Files\mosquitto

   (c) Then execute this line to execute mosquitto.conf, which should initialize the Eclipse Mosquitto broker:

   `mosquitto -v -c mosquitto.conf`

   Now, you can see the initialization of the broker in the command prompt.



   You can also execute mosquitto.config file straight away, but you will not see what events happen in it.

   (d) Now open another Command Prompt and the following command to encrypt J.Lo's password:

   `mosquitto_passwd -U pwfile.example`

   (e) For closing Mosquitto, press Ctrl+C (for Windows) on the Command Prompt, or just close it.

   Procedures for a particular error of "Unable to create websockets listener on port 9001", To fix this condition, open Task Manager as administrator and close Mosquitto. For further details, please see [McroChip1].

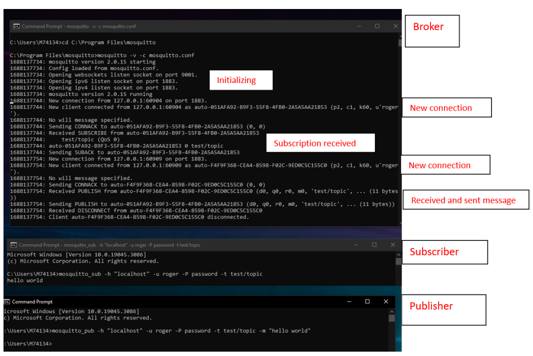
5. Testing Mosquitto

   (a) Open a command prompt and initialize mosquito, as explained in a previous section. Then, execute this line in the command prompt: `mosquitto_sub -h "localhost" -u roger -P password -t test/topic`

(b) Now, you have a command prompt that is a client subscribed to test/topic. The explanation for that line is:

	Command	Explanation
	mosquitto_sub	command for subscribing to the Mosquitto broker
	-h "localhost"	-h means the next word is the IP of the Mosquitto broker. If we write "localhost" instead, it means the IP address is the same as the computer where we are executing the line
	-u roger	roger is the username. This part is not strictly needed
	-P password	password is the used password. This part is not strictly needed
	-t test/topic	choose the topic you want to subscribe to

6. Now, open another command prompt, and execute this line: `mosquitto_pub -h ‘‘localhost’’ -u roger -P password -t test/topic -m "hello world"` The structure of the line is the same. This should make the other command read the message that this command is publishing. See the accompanying image showing what it should look like.



Step 2: General Procedure for the programming / configuring the AVR-IoT Boards (Both Board A (Publish) and Board B (Subscribe)) The first task is to add the MQTT library into MPLAB® Code Configurator (MCC) [MPLAB1]

Step 2.1: Addition of MQTT Library for MPLAB Code Configurator (MCC)

1. Download the Library

   (a) Download the Library MQTT Library for MCC MQTT Library.

2. Import the above downloaded MQTT library files into MPLAB X IDE: (If you are unfamiliar with the process, read the article How to Add a Library in MCC Import MPLAB).

3. Create a new project for AVR-IoT WG and open MCC. In Device Resources, click the Green Plus Icon next to MQTT.

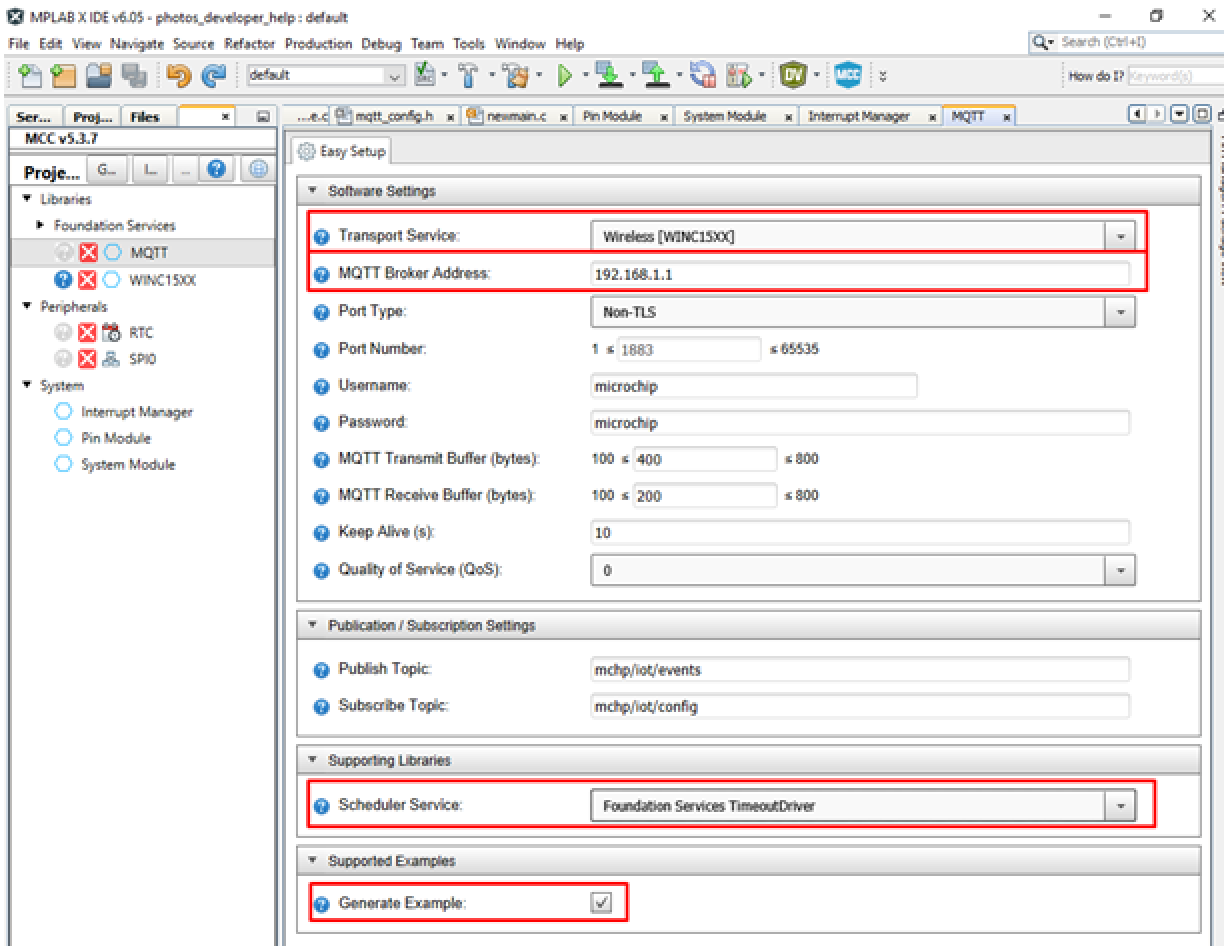
4. The MQTT Foundation service should be added. When you change the Transport Service from Custom Service to Wireless, the WINC15XX and SPI0 peripheral libraries should appear. WINC1510 is connected to the microcontroller ATmega4808 in AVR-IoT WG.

5. The AVR-IoT WG board AVR-IOT-WG contains the WINC1510 wireless module and the ATmega4808 microcontroller. ATmega4808 microcontroller is programmed, but it is not capable of connecting to Wi-Fi® by itself. It connects to the WINC1510 using SPI, and WINC1510 connects to the Wi-Fi.

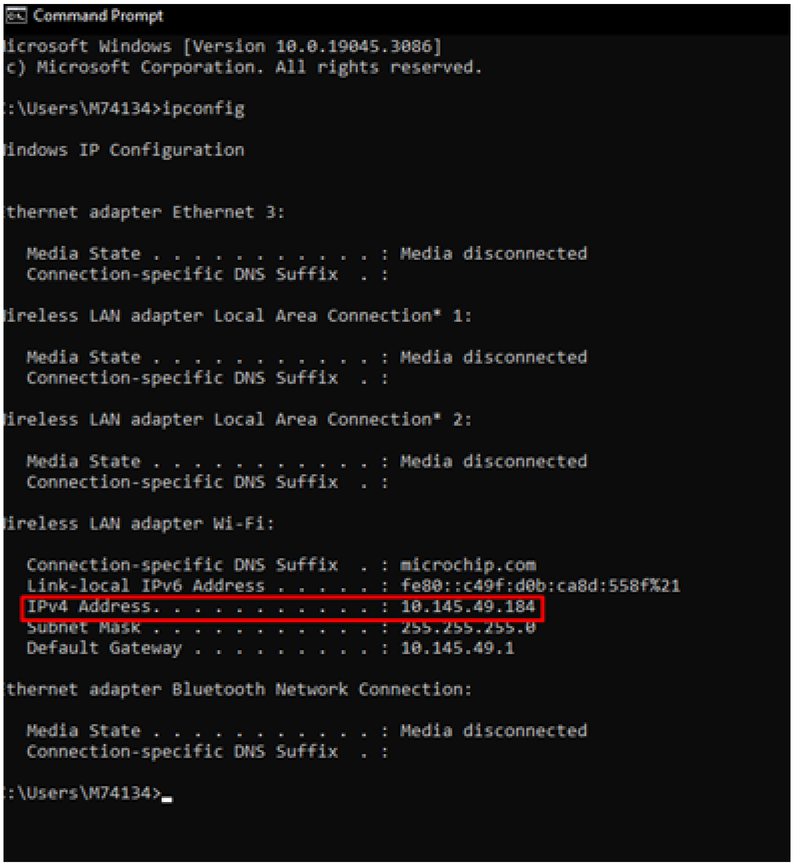
Step 2.2 Configuring MQTT Window Settings in the MQTT window in MCC

	Setting	Explanation
	Transport Service	Selects how MQTT is going to be used (Wired or Wireless)
	MQTT Broker Address	This is the IP, "the name" of the broker device that we are going to connect to
	Publish Topic	mchp/iot/events (Default - You can change) [In our experiment, we are going to have Board B does receive the published message from default topic (published by Board A) by subscribing to that default topic, while simultaneously publishing some other topic, which we don't monitor. Board A only publishes on a default topic].
	Scheduler Service	This setting selects, what tool will be used to measure time to make periodic publishing. Generate Example should be checked (This would generate default files for MQTT)

As shown in the accompanying image, you just change Transport Service, MQTT Broker Address, Scheduler Service, and tick the Generate Example box. Set these settings as in the image below except for the MQTT Broker Address, which is explained after the image. Changing the Scheduler Service to Foundation Services Timeout for the first time should make the RTC peripheral appear. The rest of the settings do not need to be changed, even if you can get a more personalized program.



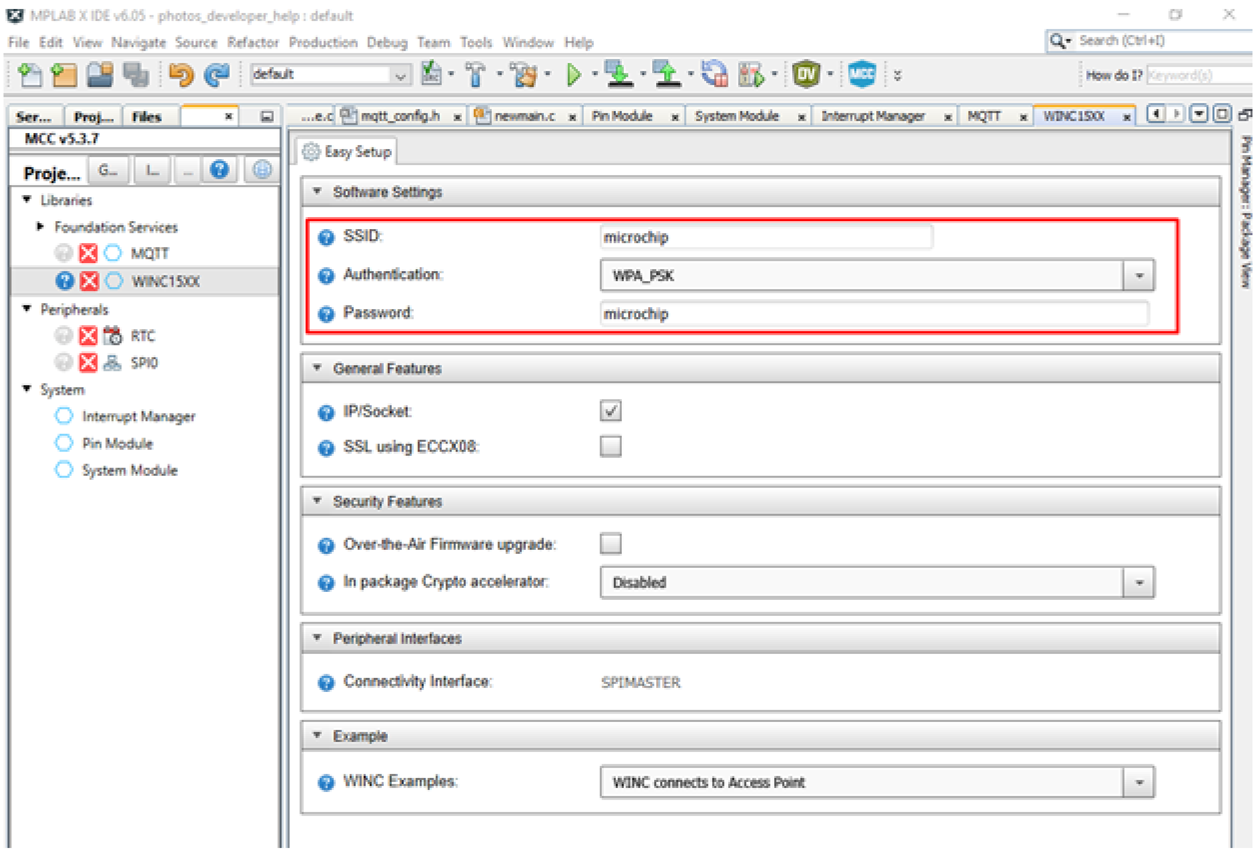
MQTT broker address is the IP address of your laptop (in which mosquitto software is loaded). To know its address, open the command prompt, and type ipconfig. The necessary address is contained in the IPv4 address's digits.



Step 2.3: Configuring WINC1510 Module in WIN15XX Software Settings Window
Change these three settings in the WIN15XX Software Settings window, as a part of configuration settings in MCC

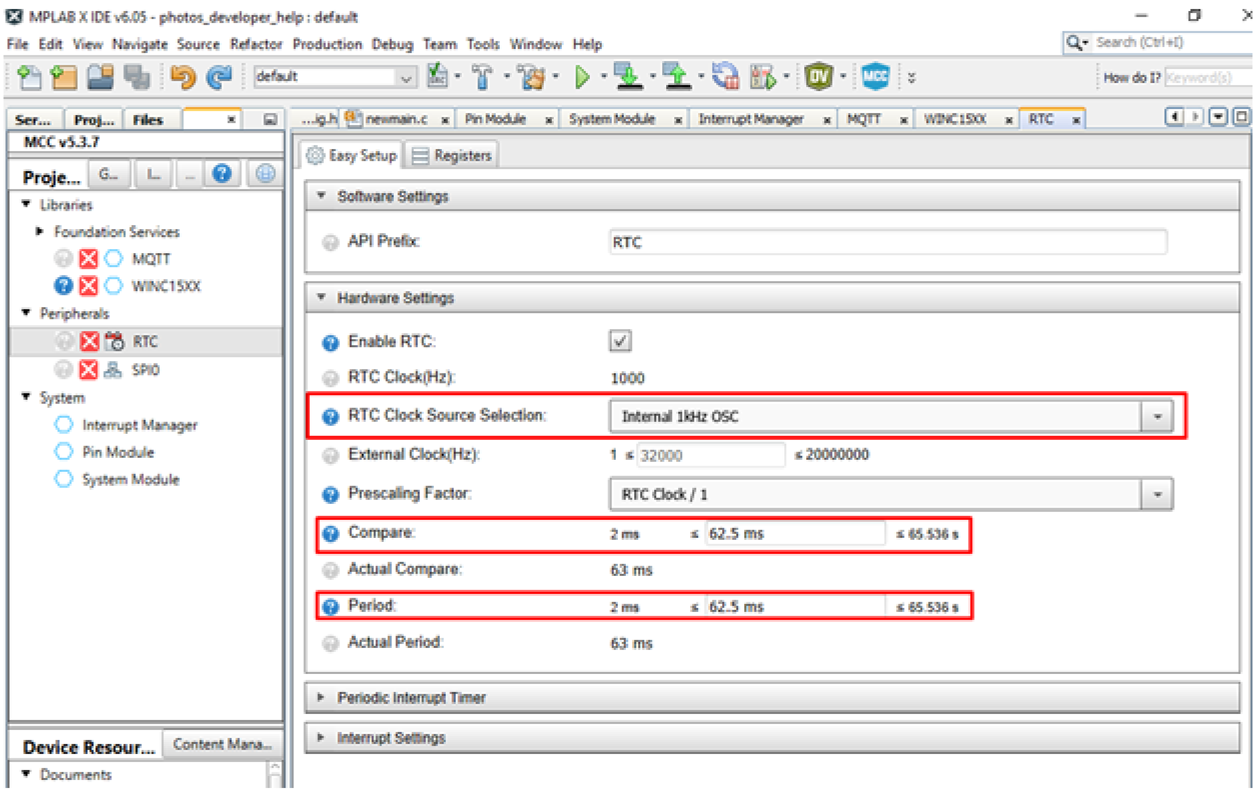
	Setting	Explanation
	SSID	Name of the Wi-Fi you are connecting to
	Authentication	Select if the Wi-Fi uses a password, WPA of WEP. Usually, your device will use WPA2. If so, select WPA_PSK
	Password	Password of the Wi-Fi you are connecting to

Change these three settings in the WIN15XX Software Settings window.



Step 2.4: RTC Window and SPI0 While you were configuring the different options, the RTC and SPI0 peripherals should have appeared. RTC is a timer used to know when to send data through MQTT, and SPI0 is the peripheral used to communicate with the WINC1510 device. If you don't see RTC and SPI0 peripherals, you may need to click on the small arrow at the left side of "Peripherals".

In this case, you won't need to change SPI0, only RTC. Select 1 kHz rate for source rate because that's how the MQTT example that you are going to generate was designed and select times for "Compare" and "Period" that are inside of the new range, between 2 ms and 65 s.



Step 2.5: PIN Module As previously mentioned, the microcontroller communicates with the ATWINC1510 W-Fi module through the Serial Peripheral Interface (SPI) protocol, using the SPI0 peripheral. SPI protocol uses 4 wires; nCS, MOSI, MISO, and SCK. All these wires are properly configured except nCS, which you need to add. nCS is connected to the PA7 pin. Although it is not needed in this exercise, more information about the SPI communication protocol can be found on the Microchip SPI Peripheral page.

To communicate with the MCU, the WINC1510 device also has the RST pin for resetting, INT pin for interrupts, EN for enabling, and WAKE pin for waking up. These pins are used for the MCU and are connected to pin PA1, pin PF2, pin PF3, and pin PF4, respectively.

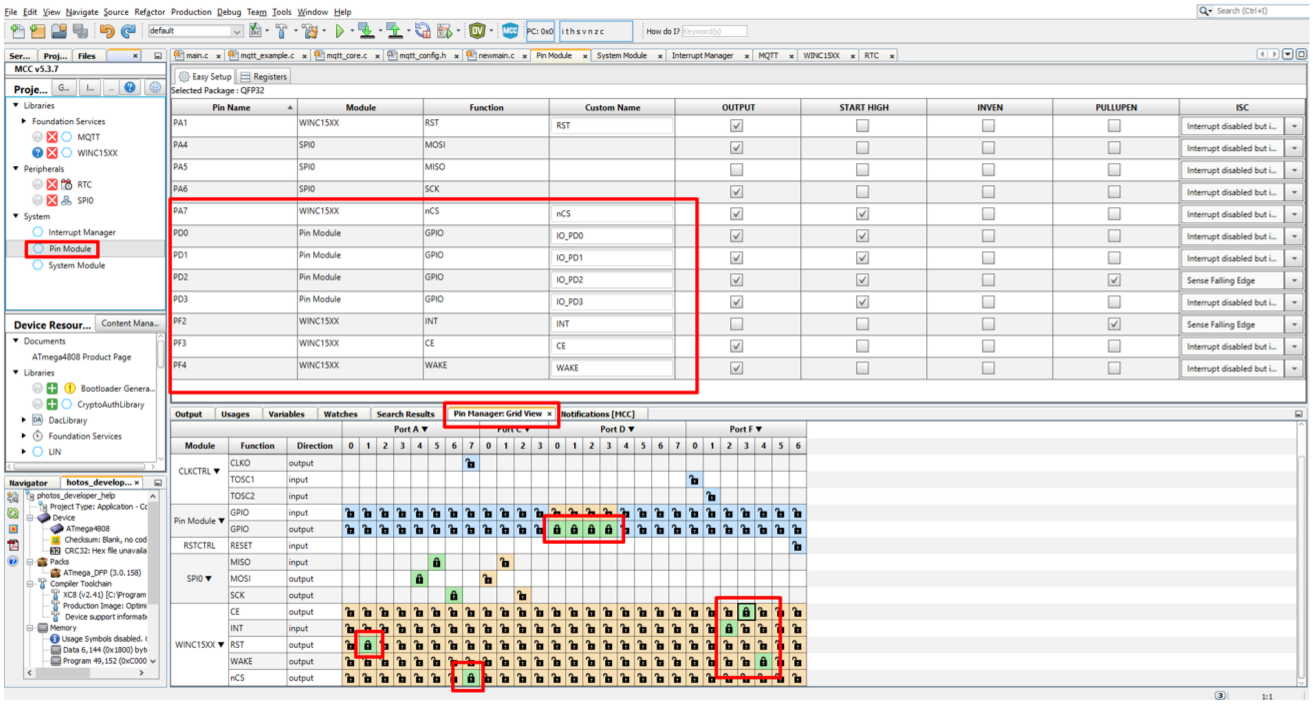
Apart from nCS, RST, INT, EN, and WAKE pins, we will also use LEDs. LEDs are connected to PD0, PD1, PD2 and PD3.

The schematic is not necessary for this example but for a better understanding of the connections, you can review it below or download the AVR® IoT WG schematic from the Microchip website.

We need to tell the MCU that we are going to use these eight pins: RST, INT, nCS and WAKE and D0, D1, D2 and D3 as GPIO output. Click in Pin Manager: Grid View and add the nine pins that we need. While you are selecting, you should see how pins are being added if you click Pin Module.

It is recommended to make D0, D1, D2, and D3 start high, by ticking the respective box.

Your final pin setup should look like the accompanying image:



Step 2.6: Generate Code If every thing goes fine, you would not see a window: Ref [MPLAB1]

If you see it, it's likely because a step was missed. To see what step may have been missed, you can click Notifications [MCC]. The notifications that you have that are not in the accompanying image, could be the steps that you are missing. Refer [MPLAB1] for more information.

Step 2.7: Write Code in main.c Write your own main() code, which would adopt the most important key MQTT library function `app_mqttScheduler()` among others.

In this lab experiment, we have written two main() programs, mainPublishCmmntd.c and mainSubscribeCmmntd.c corresponding to source and sink nodes respectively, though mandatorily every IoT under MQTT is both a publisher & subscriber. (But, we can mute one of functions - either publish or subscribe, given an IoT - which is the case for majority of the engineering requirements).

Apart from this, two support files M2M-MQTT-MosquittoPblshM-Cmmntd.c and M2M-MQTT-MosquittoSbscrbeM-Cmmntd.c corresponding to publish and subscribe nodes. You are supposed to replace the example.c code corresponding to MQTT protocol, with the above codes, in the IPE MPLAB, compile it and burn in, the respective AVR-IoT boards. [For task 2, you need to incorporate ADC in publish board and LCD interfacing in the subscribe board, which is taken care in the support files. These extra tasks are detailed in the next section].

Step 3: Writing the ADC for light sensor data & LCD Interfacing (Task 2)

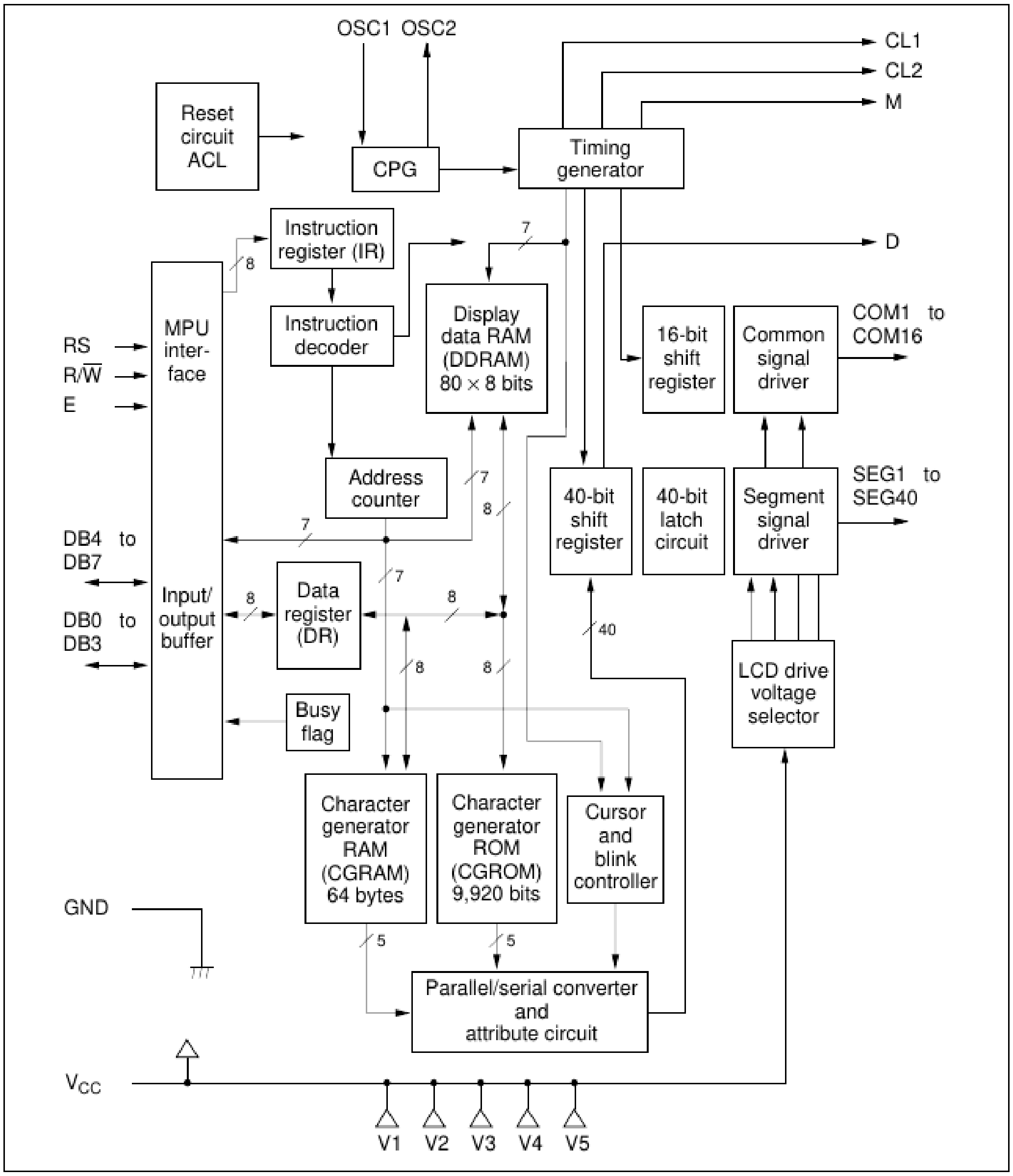
Step 3.1: Writing the ADC for Light Sensor Data in the AVR-IoT Publish Board ADC of light sensor data has to be done only in the publish node. 10-bit ADC is used for conversion.

The code for the ADC is done in the mainPublishLuxADC.c itself (and hence not incorporated in publish support files). Note that there are blanks and you need to fill-in, show it to the TAs and then continue (burn and run) the experiment.

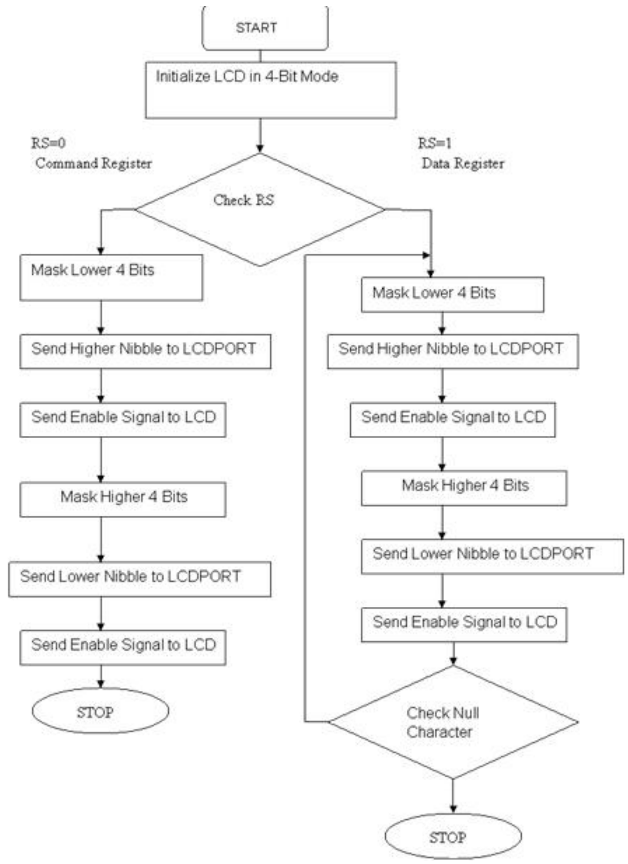
Step 3.2: LCD Interfacing with the AVR-IoT Subscribe Board (Task 2) Following is the connections of LCD with AVR-IoT board.

AVR Microcontroller Pin	LCD Screen Pin	Description
PA2	RS	Control Pin (Register Select)
Ground	RW	Control Pin (Read/Write, connect to Ground)
PA3	E	Control Pin (Enable)
PC0	D4	Data Pin (4-bit mode)
PC1	D5	Data Pin (4-bit mode)
PD6	D6	Data Pin (4-bit mode)
PD4	D7	Data Pin (4-bit mode)
+5V	VDD	Power Supply (5V)
Ground	VSS	Ground
Potentiometer Wiper	V0/VEE	Contrast Adjustment
+5V	Potentiometer VCC	Power Supply (5V) for Potentiometer
Ground	Potentiometer Ground	Ground for Potentiometer

Block diagram for the LCD that we are using in the lab (HD44780U), is given below, though it is not necessary, yet given here for academic reasons.



LCD displaying algorithm is given below in terms of flow chart



LCD is to be initialized, for displaying data in 4-bit mode. Other configurations like cursor ON/OFF, clear display, by writing a series of configuration bits/bytes, as given in the code M2M-MQTT-MosquittoSubscribe.c. All the interfacing LCD algorithms are implemented in this file itself with some LCD parameters set in the mainSubscribe.c.

Step 3: Burning the Object Code into the Boards In this step, we would edit the base / example codes (one for Board A (Publish) and another for Board B (Subscribe)) in MPLAB. This tweaked / edited file being supportPublish.c, to copy in place of /Source Files/MCC_Generated Files/examples/mqtt_eample.c file in the same directory in MPLAB environment [MPLAB1]. Burn the code [MPLAB1]. Go through the user guide AVR-IoT-WG of AVR-IoT WG board [AVR-IOT-WG-UsrGde] to know how the hardware connections to be made, during burning the code.

Step 3.1: Burning Code in Board A (Publish) The code (mainPublish.c) along with the suitable support files (SupportPublish.c) would only publish and is meant for Board A (Publish). Download these codes from moodle, repeat the procedure given in Step 2 (import, configure in MPLAB in your laptop, replace with SupportPublish.c, and so on). Connect the Board A to your laptop through USB and click the "*run main project*" in the MPLAB window to burn the code into Board A (Publish).

Step 3.2: Burning Code in Board B (Subscribe) The code for Board B (Subscribe) consists of two parts: (a) the subscribe part of MQTT protocol (and there is also a publish part, inherently comes along) and (b) the received message through subscription has to be displayed in the LCD (i) Hello message and (ii) light sensor data.

Part (a) Similar to the publish node, code (mainSubscribe.c) and suitable support files (supportSubscribe.c) are different, in which the Board B (Subscribe) does receive the published message from default topic, by subscribing to the default topic, while simultaneously publishing some other topic (EE2016IITM), which we don't monitor. Hence, in Configuring MQTT Window need to choose the right topic for both publishing and subscribing. In case, you shut down the mosquitto server, you need to reset the boards. Run the Mosquitto server and view the results.

Part (b): We use the 16 x 2 LCD (Hitachi See HitchiLCD) to display the message received through subscription. Refer [AVR-IOT-WG-UsrGde] to the PIN connections of AVR-IoT WG, and the details given in section 3.1 and accordingly make the connections for the LCD to AVT-IoT WG board terminals.

5 Tasks

1. (Task 0) Do the experiment in your hostel room itself by configuring your laptop as a MOSQUITTO broker and your mobile & your friend's mobile as a publish and subscribe nodes.

2. (Task 1) Change the Publish rate and observe the corresponding received rate through Subscription.

   (a) Go to (in MPLAB) main()–>header files –> MCC Generated files –> config –> MQTT_config.h CFG_MQTT_CONN_TIMEOUT 10 (as generated by MPLAB - you can change the publish rate to 6, 10, 14, 18 & 22 - & run for each time. Observe the LED blink rate for each case).

3. (Task 2a) Receive, 'Hello', message & display it. Independently transmit to display Hello in LCD. Now, interface it with MQTT iot protocol. The problem is that the MQTT uses many of the pins in each available ports (PORT A, C & D). Seems you need to take some pins from a port and remaining from another port and combine them to send nibble at a time to the display LCD.

4. (Task 2b) Receive the light sensor data and display it.

Acknowledgements: The author acknowledges, Yuvasri and Prasanna to take up this as a summer internship project 2024, and working on it on a daily basis, Saran Prithive for implementation of software aspects, Anand for fixing the LCD interfacing part, apart from Bhuvaneshwari for management.

References

[McroChip1] https://developerhelp.microchip.com/xwiki/bin/view/products/wireless-connectivity/Embedded-Wi-Fi/mqtt-in-mplab/mqtt-environment/

[Msqtto] https://mosquitto.org/download/

[MPLAB1] https://developerhelp.microchip.com/xwiki/bin/view/products/wireless-connectivity/Embedded-Wi-Fi/mqtt-in-mplab/first-steps-in-mplab/

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